Clim Dyn
DOI 10.1007/s00382-016-3092-9

Investigating the zonal wind response to SST warming using transient ensemble AGCM experiments

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Received: 30 March 2015 / Accepted: 17 March 2016
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Abstract The response of the atmospheric circulation to greenhouse gas-induced SST warming is investigated using large ensemble experiments with two AGCMs, with a focus on the robust feature of the poleward shift of the eddy driven jet. In these experiments, large ensembles of simulations are conducted by abruptly switching the SST forcing on from January 1st to focus on the wintertime circulation adjustment. A hybrid, finite amplitude wave activity budget analysis is performed to elucidate the nonlinear and irreversible aspects of the eddy-mean flow interaction during the adjustment of the zonal wind towards a poleward shifted state. The results confirm the results from earlier more idealized studies, particularly the importance of reduced dissipation of wave activity, in which the midlatitude decrease of effective diffusivity appears to be dominant. This reduction in dissipation increases the survival of midlatitude waves. These surviving waves, when reaching the upper propagation level in the upper troposphere, are subject to the influence of the increase of reflection phase speed at the poleward side of the mean jet, and thus more waves are reflected equatorward across the jet, giving rise to a poleward transport of momentum and thus an eddy momentum flux convergence for the poleward shift. The relative importance of wave breaking-induced PV mixing versus diabatic PV source in the evolution of the Lagrangian PV gradient is also investigated. The former plays the dominant role in the PV gradient formation during the initial phase of the jet shift, while the latter actually opposes the evolution of the Lagrangian PV gradient at times.

Keywords Greenhouse warming · Finite amplitude wave activity · Wave reflection · Phase speed · Effective diffusivity

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1 Introduction and motivation

Poleward displacement of the major overturning circulation cells and a corresponding shift of the eddy-driven westerly jet have been identified as robust features of the response of the global atmospheric circulation under greenhouse warming (e.g., Lu et al. 2007; Staten et al. 2012). Given the close relationship between these circulation changes and the global pattern of the hydrological cycle and related extremes (e.g., Seager et al. 2007; Lorenz and DeWeaver 2007a; Scheff and Frierson 2012; Lu et al. 2014), considerable efforts have been made to understand the underlying mechanisms of the jet shift. It has been attributed to

- the increase of the subtropical and midlatitude static stability (Lu et al. 2008, 2010);
- the rise of the extratropical tropopause (Lorenz and DeWeaver 2007a);
- the increase in the length scale of the midlatitude waves (Williams 2006) and the resulting reduction of the relative phase speed of the waves with respect to the background zonal wind (Kidston et al. 2011);

Published online: 13 April 2016

- the increase in the upper level baroclinicity (Yin 2005; Rivière 2011), and the consequent increase in the eddy length scales;
- the increase of the phase speed of the equatorward propagating Rossby waves and hence the poleward displacement of their critical latitudes, due to the increase of the meridional temperature gradient near the tropopause in the extratropics reducing the background gradient of the potential vorticity (PV) and hence the intrinsic westward phase speed of the Rossby waves (Chen et al. 2007, 2008; Wittman et al. 2007);
- the enhanced reflection (Lorenz 2014b) or refraction (Wu et al. 2012) of the wave activity at the poleward side of the jet by the increased upper tropospheric zonal wind; and
- the decrease and increase of effective diffusivity at the poleward flank of the jet stream and in the subtropics, respectively, with the former reducing the dissipation of wave activity and allowing more waves joining the equatorward propagation in the upper troposphere and the latter enhancing the subtropical mixing of PV and pushing the PV gradient maximum and hence the jet poleward (Chen et al. 2013; Lu et al. 2014).

The diversity of the mechanisms described by these studies underscores the complexity of the midlatitude circulation response to global warming.

While the majority of the aforementioned studies points to an extratropical origin for the jet shift, emerging evidence suggests that changes in the tropical troposphere can also give rise to a poleward shift of the eddy-driven jet through robust dynamical processes. Nearly inviscid theory (Schneider 1977; Held and Hou 1980) suggests that in an axially symmetric atmosphere, the jets could be displaced poleward if the radiative-convective equilibrium temperature contrast between the tropics and extratropics was increased, or if the tropical tropopause was raised. Schneider (1984) demonstrated, however, that jet position in the nearly inviscid axially symmetric limit is equatorward of its observed position, and that the observed position is in some sense a balance between tropical heating pulling it equatorward towards the axially symmetric position via the axially symmetric mechanism, and eddy momentum fluxes pulling it poleward.

Butler et al. (2011) first showed that a broad tropical heating mimicking the tropical upper-tropospheric warming in fully coupled climate models can induce an anomalous PV gradient at the edge of the warming, and that the resultant down-gradient PV flux provides the eddy forcing for the poleward shift of the jet. The robustness of this tropical heating in producing a poleward jet shift was verified by Sun et al. (2013) and Lu et al. (2014) using a different dry dynamical model. The latter two studies further demonstrated that both narrow and broad tropical upper-tropospheric heating can produce qualitatively similar extratropical response (different in magnitude) as if the response is projected onto the same attractor of the dynamical system irrespective of the width the imposed heating. A likely rationalization for this insensitivity is the mediating effect of the nearly inviscid zonally symmetric response of the tropics to the thermal forcing, which always puts the forced tropospheric temperature gradient at the edge of the tropics.

Lu et al. (2014) further showed that the eddy response to a tropical upper tropospheric thermal forcing does not follow what the linear wave refraction analysis would predict, but is dominated by the irreversible eddy mixing of the PV involving the evolution of the effective diffusivity itself (see also Chen et al. 2013). In their case, the linear wave refraction functions as a positive feedback to, but not an initiator of, the change of the background zonal wind. Lu et al. (2014) and Chen et al. (2013) adopted an Eulerian–Lagrangian diagnostics of finite-amplitude wave activity (FAWA, Nakamura 1996; Nakamura and Zhu 2010) to diagnose the anomalous EP-flux divergence during the different stages of the transient eddy adjustment. This finite-amplitude wave activity diagnostic is unique in that it is the result of a coordinate transformation from the traditional Eulerian coordinate to a tracer-following coordinate, specifically, the equivalent latitude. The resultant wave activity is positive definite and the related non-acceleration theorem is satisfied without approximation for adiabatic and frictionless flow. This framework aids in focusing the analysis onto the irreversible processes of the eddy adjustment. Applying the FAWA budget (see Sec. 2.3 in Nakamura 1996 for details), Chen et al. (2013) and Lu et al. (2014) identified several key processes that are important in leading up to the poleward shift of the eddy-driven jet. By diagnosing the change of effective diffusivity, they found that a tropical upper tropospheric heating first drives an enhanced equator-to-pole temperature gradient and enhances the PV gradient at the edge of the Hadley cell (through the nearly inviscid, axially symmetric adjustment in the absence of eddy feedback). As a consequence, the effective diffusivity there is reduced and hence the wave activity coming from below is less strongly dissipated and can reach the free propagation layer in the upper troposphere (around 200–250 hPa levels). As waves in this layer propagate predominantly equatorward, the more wave activity survived there results in more wave breaking and dissipation near the subtropical critical latitude. This process is represented by an increase of effective diffusivity in the subtropics in the wave activity diagnostic. The associated PV structure during the adjustment bears some resemblance to that found in Butler et al. (2011). However, the anomalous PV flux is not down the anomalous PV gradient as found in their work.

(Down-gradient PV flux implicitly implies that diffusivity remains unchanged.) Instead, the dipolar PV anomalies, as well as the structure of the anomalous effective diffusivity, are not fixed in space, but translating poleward with time until they reach their final equilibrated positions, consistent with the shifted jet. Then the irreversible mixing of PV and dissipation of wave activity—the inevitable consequence of nonlinear wave breaking—is essential for the creation of the new PV gradient and the shaping of the final PV structure.

Recently, Lorenz (2014a, b, 2015) developed a Rossby wave chromatography based on a linearized barotropic model and used it to diagnose the barotropic interaction between the mid-latitude jet and eddies. He found that change in the 'reflective level' on the poleward flank of the jet is crucial for the poleward shift of the jet in response to external forcing. Staten et al. (2014) examined the reflection phase speed in the ensemble simulations with GFDL AM2.1 and found that indeed when the zonal wind increases on the poleward flank of the jet under the SST forcing, a wider range of poleward propagating waves encounter a reflection level instead of a critical level on the poleward flank. The increased wave reflection leads to increased equatorward wave activity propagation across the jet, resulting in a more poleward jet. Rather than challenging the RWC perspective, the present study reconciles the equatorward wave reflection with the structure of the nonlinear wave dissipation diagnosed from the wave activity budget analysis.

The main purpose of the present study is to examine (1) whether the notion about the role of irreversible mixing in the jet shift gained from the idealized AGCM experiments applies to the northern hemisphere (NH) zonal wind response in a more realistic setting in real world AGCM simulations; (2) whether the paradigm suggested by Lu et al. (2014) and Chen et al. (2013) or that suggested by Butler et al. (2011) is more relevant to the NH midlatitude response; and (3) the dynamical processes that are responsible for the structure of the changes in wave dissipation. To this end, we conduct similar large ensemble transient experiments to Chen et al. (2013) and Lu et al. (2014), but using two comprehensive AGCMs—NCAR CAM4 and GFDL's AM2.1. We will perform the similar FAWA analysis as in Chen et al. (2013) and Lu et al. (2014) to assess to what extent the processes identified in the idealized models are also operating in the comprehensive AGCMs with realistic zonal asymmetry. In addition, the FAWA analysis here is extended to include the contribution from the diabatic sources to the FAWA budget, since Chen et al. (2013) and Lu et al. (2014) ignored the diabatic wave source in their analysis.

The rest of the paper is arranged as follows. We introduce model experiments and the diagnostic methods in Sect. 2. In Sect. 3 we discuss the general results from the equilibrium and transient simulations. In Sect. 4 we analyze the dynamical processes leading to the jet shift. In this section, we also discuss the sources of the PV gradient, analyze how the background PV gradient is created during the transient adjustment, and compare the adjustment in the Southern Hemisphere (SH) with that in the NH.

2 Data and methodology

2.1 Modeling strategy

The sea surface temperature (SST) pattern due to global warming consists of features such as polar amplification (e.g., Manabe et al. 1990; Holland and Bitz 2003), enhanced equatorial warming (Liu et al. 2005), an enhanced and shifted midlatitude gradient (Bengtsson et al. 2006), and of course the warming itself. It is thought that the global mean warming component of the greenhouse gas-induced warming is of prime importance for the shift of the midlatitude atmospheric circulation in response to global warming. Therefore, we perform large ensemble of 121 day simulations forced by an abrupt increase of SST by 4°K from Jan. 1, using the CAM4 (Community Climate System version 4) AGCM of the National Center for Atmospheric Research (NCAR). A similar set of transient ensemble simulations was also performed by Staten et al. (2013, 2014), but using the full SST anomaly pattern at the end of the twenty-first century in the A1B scenario as simulated by the GFDL coupled GCM CM2.1. Comparison between the large ensemble simulations using the two AGCMs and with two different SST perturbations will indicate the robustness of the results. The main benefit of the large ensemble experiments is to reveal the "signal" of the dynamical adjustment to the abrupt switch-on of the forcing, since the noise is greatly reduced by the ensemble averaging. This makes it possible to discern the important dynamical processes leading to the final circulation response, and to test proposed cause-effect relationships.

2.2 CAM4 versus AM2.1

The two AGCMs used for the purpose of this study are CAM4 (Gent et al. 2011) and AM2.1, the atmospheric component of CM2.1 (Delworth et al. 2006). CAM4 is the version of the community atmospheric model used for the CMIP5 project (Taylor et al. 2012) by NCAR and represents a significant improvement from the earlier version, while CM2.1 and AM2.1 were used for CMIP3 (Meehl et al. 2007). It is especially noteworthy that the modification of the deep convection parameterization in CAM4 from its Zhang and McFarlane (1995) predecessor in

CAM3 by including more realistic dilution effects through an explicit representation of entrainment, as well as by the inclusion of cumulus momentum transport, has significantly improved the spatial and temporal characteristics of the simulated ENSO variability in the associated CESM CGCM and the associated teleconnections (Neale et al 2010), possibly lending credence to this model's response to global warming-related forcing. Moist convection in AM2.1 is represented by the relaxed Arakawa-Schubert (RAS) formulation of Moorthi and Suarez (1992). Both models use a finite volume dynamical core with a nominal 2° latitude × 2.5° longitude horizontal resolution. CAM4 has more vertical levels (26) than AM2.1 (24), with finer resolution near the tropopause and lower stratosphere.

For each model, two types of experiments were performed: one type consisted of equilibrium runs with a repeating seasonal cycle of SSTs and sea ice integrated for many years, and the other, large ensembles of short transient simulations. For CAM4, we conducted a 50-year long control simulation with the current (representing year 2000) seasonally varying SST and sea ice climatology boundary conditions, a 50-year long perturbation run with a 4°K uniform warming added to the control SST, and a pair of switch-on experiments. For the control switch-on ensemble, the model branches off from January 1st with the same SST as the equilibrium control but 320 different initial atmospheric conditions, thus constituting an ensemble of 320 members, each being integrated for 122 days. This control ensemble will be referred to as CAM4_CNTRL ensemble. The corresponding perturbation ensemble (will be referred to as CAM4_4K) was similarly performed but with a 4°K uniform SST perturbation added to the control SST boundary condition. Sea ice condition was not perturbed in CAM4_4K ensemble.

Similar suites of experiments were also conducted with AM2.1 and the results of the earlier analysis were reported in (Staten et al. 2012, 2014). For the present study, we make use of a 200-year long control equilibrium simulation, the corresponding 200-year long SST perturbation simulation, and a pair of 640-member ensembles of SST switch-on experiments (each member is 90 days long). The SST data for the control experiment is calculated from a coupled pre-industrial control (representing year 1900 climate) simulation. The forcing for the SST perturbation simulation is the ten-year averaged (from 2091 to 2100) SST and sea ice from a coupled A1B scenario simulated by the CM2.1 model. Unlike the 4°K uniform warming used in the CAM4 experiments, here the prescribed SST perturbations vary by longitude, latitude and month. The AM2.1 control ensemble will be referred to as AM21_CNTRL and the SST perturbation ensemble AM21_A1B, The diabatic terms were not archived for the AM2.1 simulations; thus, we will not be able to consider their contribution to the wave activity budget in Sect. 4.

2.3 Finite amplitude wave activity

The main diagnostic tool for understanding the eddy adjustment to SST forcing is a hybrid Euler-Lagrangian finite-amplitude wave activity budget developed by Nakamura (1995, 1996). It has been implemented for the wave activity adjustment in both dry and moist idealized AGCM simulations (e.g., Nakamura and Solomon 2010; Chen et al. 2013; Sun et al. 2013; Lu et al. 2014). The detailed development of the concept of finite-amplitude wave activity and its budget for a quasi-geostrophic atmosphere can be found in Nakamura (1995, 1996) and Nakamura and Zhu (2010). The diagnostic framework is described below.

The FAWA equation can be considered to be the area-coordinate transformation of the familiar quasi-geostrophic potential vorticity (QG PV, or PV in shorthand) equation that governs the midlatitude dynamics. It can be written as

$$\frac{\partial}{\partial t}q + J(\psi, q) = k\nabla^2(q - f_0 - \beta y) + \vartheta \tag{1}$$

where q is the potential vorticity, ψ is the horizontal streamfunction, f_0 is the reference planetary vorticity, β is $\frac{2\Omega\cos\phi}{a}$, J is the horizontal Jacobian, k is the diffusion coefficient and ϑ is the QG PV source due to the diabatic processes such as radiation and condensation and is expressed as $\vartheta = f\left(Q_r/_{\theta_{0p}}\right)_p$ (see Eq. 3.2.14 in Andrews et al. 1987), where f is Coriolis parameter, Q_r is diabatic heating rate, θ_{0p} is the horizontally averaged potential temperature at a constant pressure.

Instead of characterizing an eddy perturbation as a deviation from a zonal average, the eddy activity measures the net areal displacement of the PV contours from zonal symmetry, and is defined geometrically following the material surface of the QG PV as

$$A = \frac{1}{2\pi a\cos\phi_e}\left[\iint_{q>Q;\phi<\pi/2} qdS - \iint_{\phi_e<\phi<\pi/2} qdS\right]. \tag{2}$$

Equation (2) describes an area-coordinate transformation from q to A. In Fig. 1, a snapshot of PV on 315 hPa isobaric level, A corresponding to PV = 2×10^{-4} s^{-1} describes the difference between integration of q within the red areas equatorward of equivalent latitude (ϕ_e) and the integration of q within the blue areas poleward of equivalent latitude (ϕ_e). The corresponding equivalent latitude (grey dashed line) to $Q = 2\times10^{-4}$ s^{-1} is determined by the requirement that the area enclosed by the Q contour towards the polar cap equals the area poleward of latitude ϕ_e:

$$\iint_{q>Q;\phi<\pi/2} dS = \iint_{\phi_e<\phi<\pi/2} dS = 2\pi a^2(1 - \sin\phi_e). \tag{3}$$

Upper case Q is used to emphasize the Lagrangian nature of a wavy, instantaneous PV contour. It has been shown that with the wave activity defined above, the

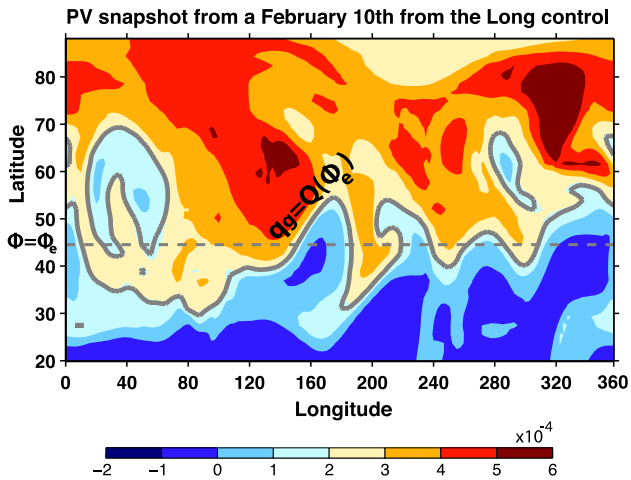

Fig. 1 Longitude-latitude snapshot of PV field at 315 hPa on a February 10th from the long control (unit: s^{-1}). The *gray contour* denotes the PV value corresponding to the equivalent latitude highlighted by the *dashed line*. The area enclosed by the *gray contour* toward the polar cap equals the area poleward of the *dashed line*. The *contorted curve* is the contour of $q_g = Q(\phi_e)$ and the *dashed curve* is the latitude line $\phi = \phi_e$. See text for details

non-acceleration theorem holds exactly with no approximation for both small amplitude and finite amplitude eddies for inviscid, adiabatic flow. Moreover, by applying the integration operation (2) to the terms of Eq. (1), one yields a hybrid Eulerian–Lagrangian FAWA budget equation,

$$\frac{\partial A}{\partial t} + \overline{v'q'} = -\frac{K_{eff}}{a}\frac{\partial Q}{\partial \phi_e} + k\frac{\partial \bar{q}}{\partial y} + \Delta\Sigma, \tag{4}$$

with the irreversible dissipation of wave activity across the PV contour being expressed as a diffusion down the gradient of PV. The $\overline{v'q'}$ is the zonal mean eddy PV flux, with the same meaning as it is in the zonal mean zonal momentum equation (thus representing the Eulerian component of the budget equation). K_{eff} is the effective diffusivity of the irreversible mixing of PV across material surfaces and, as shown by Nakamura (1996), it takes the form $K_{eff} = \frac{(k|\nabla q|^2)}{\left(\frac{\partial Q}{\partial y}\right)^2}$ where, k is the diffusion coefficient for regular diffusion and $\frac{|\nabla q|^2}{\frac{\partial Q}{\partial y}^2}$ can be interpreted as a 'normalized equivalent length' of the PV contours, measuring the enhancement to diffusivity from the advective stretching, a process that not only elongates the interfaces between different PV materials, but also enhances the cross-stream gradient of PV (both enhancing the diffusivity). $\Delta\Sigma$ is the source of wave activity due to diabatic heating/cooling and can be calculated through the area-coordinate transformation applied to ϑ

$$\Delta\Sigma = \frac{1}{2\pi a \cos\phi_e}\left[\iint_{q>Q;\phi<\pi/2} \vartheta\, dS - \iint_{\phi_e<\phi<\pi/2} \vartheta\, dS\right]. \tag{5}$$

As k was found to be an order of magnitude smaller than K_{eff} (while $\frac{\partial \bar{q}}{\partial y}$ is of the same order as $\frac{\partial Q}{\partial y}$), the second term on the right hand side is ignored in the following budget analysis.

In a full AGCM, k represents numerical diffusion as well as the diffusive effects from other sub-grid processes (such as gravity waves, cumulus cloud friction). Since it is not explicitly provided in the model output, the value of k is estimated as a residual assuming all the terms in (4) are well balanced. Thus, the k value in Eq. (4) should be only interpreted in a heuristic sense.

3 The equilibrium and transient responses to SST forcing

We first examine the main features of the zonal mean equilibrium response of the winter (JFM) circulation to the SST forcings in the two AGCMs. In spite of the simple uniform SST perturbation, CAM4 captures the general structures of the zonal mean zonal wind and zonal mean temperature response as simulated in CMIP3/5 models under future warming scenarios (e.g., Lorenz and DeWeaver 2007b; IPCC report 2007; Grise and Polvani 2014), as shown in Fig. 2. These include (1) the strengthening of the zonal winds at and above the cores of the subtropical jets in both hemispheres; (2) a poleward shift of the eddy-driven component of the westerlies as embodied in its barotropic structure in the troposphere; (3) an enhanced tropical upper tropospheric warming, which is in close to thermal wind balance with the corresponding changes in the zonal wind; and (4) cooling (or lack of warming) near the levels above the extratropical and equatorial tropopause. Features (1) and (2) collectively lead to a greater split between the subtropical and eddy-driven jets, a symbolic wind anomaly structure due to global warming (Lorenz and DeWeaver 2007b). All the features above are also shared by the response in AM21 forced by the A1B type SST perturbation (Fig. 3). It is somewhat surprising to see that the SH jet shift in CAM4 is relatively weaker than its NH counterpart during the same months (JFM), while during these months the CMIP5 multi-model ensemble exhibits the largest and most robust response in the SH eddy-driven wind (Grise and Polvani 2014). One possible culprit might be that the mean state of the JFM zonal wind in CAM4 is seriously biased, compared to the more realistic AM2.1 climatology. The SH wind shift in the equilibrium simulation with AM2.1 is much stronger despite of the weaker global mean SST (approximately 2°K) warming compared to the 4°K used for CAM4, indicative of the possible role of the SST gradient in driving the eddy-driven jet shift in the SH (Bengtsson et al. 2006). Given the discrepancy in the SH response between the two models, we in this study will mainly focus on the NH jet response. For the NH response, as will be demonstrated later, the dynamical processes

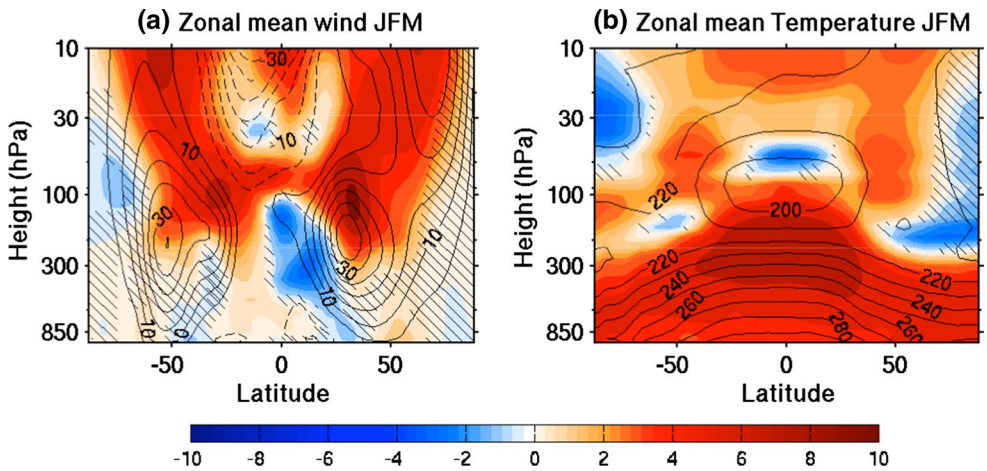

Fig. 2 Latitude-height plots of zonal mean zonal wind (**a**) and zonal temperature (**b**) responses to 4 K uniform warming during the JFM season simulated by CAM4 are *shaded*. The *overlaid contours* represent the corresponding climatological fields and the contour interval is 5 ms^{-1} for the zonal mean wind and 10 K for zonal mean temperature, respectively. *Hatched areas* are where the response does not pass the significance test at 5 % confidence level

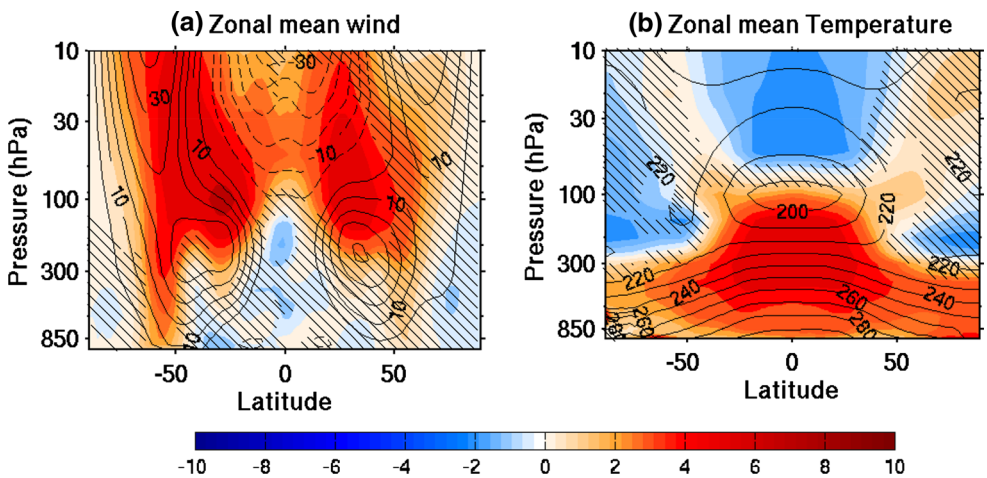

Fig. 3 Latitude-height plots of zonal mean zonal wind (**a**) and zonal temperature (**b**) responses to A1B type warming during the JFM season simulated by AM2.1 are *shaded*. The *overlaid contours* represent the corresponding climatological fields and the contour intervals are 5 ms^{-1} for the zonal mean wind and 10 K for zonal mean temperature, respectively. *Hatched areas* are where the response does not pass the significance test at 5 % confidence level

responsible for the shift of the jet are similar between the CAM4 and AM2.1 experiments. Corroborating the earlier studies (e.g., Butler et al. 2011; Chen et al. 2013), the uniform component of the SST warming and the associated upper tropospheric warming are sufficient for forcing the midlatitude circulation response seen in the full climate models. Considerable differences exist in the stratospheric response between the two models, but they are beyond the scope of the present investigation.

To focus on the eddy-driven component of the wind response, we show in Fig. 4a the transient evolution of the ensemble mean zonal mean zonal wind response at 850 hPa level to the switch-on of 4°K warming from Jan. 1 in CAM4 (i.e., CAM4_4K-CAM4_CNTRL). The 850 hPa wind response from the AM2.1 ensemble runs is shown in Fig. 4b for comparison. Both ensemble responses share some similar features in their temporal evolution in both NH and SH. In both cases and both hemispheres, the 850 hPa wind response is characterized by a tri-polar structure, indicating that the eddy-driven wind evolves towards a poleward shifted state with time. In both models, the time scale of the eddy-driven wind adjustment exhibits similar inter-hemispheric asymmetry: the tri-polar structure develops much faster in the NH (around day 25–30) than the SH. Moreover, even the wind structures prior to the formation of the jet shift at the initial stage of the adjustment show considerable similarity between the two models, indicating similar development mechanisms. Guided by the temporal characteristics of the adjustments in both hemispheres, in the following sections we focus our analysis on the linear trend during the period of days 1–25 for the NH and days 1–50 for the SH, respectively.

4 Dynamical processes for the shift of the eddy driven jet

To facilitate the comparison with the earlier idealized studies (Chen et al. 2013; Lu et al. 2014) for the mechanisms of the eddy-driven jet response to SST warming, we implement the FAWA budget analysis for the EP-flux divergence (equivalent to eddy QG PV flux) for the transient ensemble

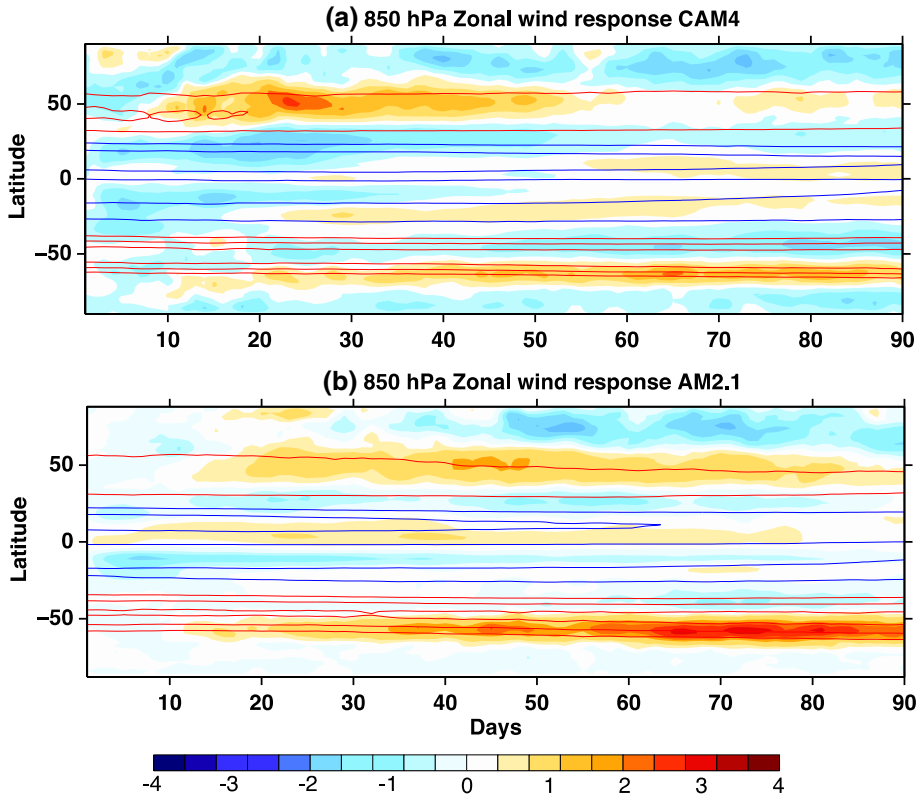

Fig. 4 Ensemble mean response of zonal mean zonal wind at 850 hPa (*shading*) to 4 K uniform SST warming in CAM4 (**a**) and to A1B SST warming in AM2.1 (**b**). Day 1 in abscissa corresponds to January 1st. *Contours* denote the climatological wind (positive contours are *red* and negative contours are *blue*) with contour intervals of 4 ms^{-1}

simulations with CAM4 and AM2.1. The budget analysis corroborates the assertions of the idealized studies regarding the importance of the irreversible PV mixing processes for understanding the mechanisms of the jet shift. On the other hand, owing to the added complexities of the full AGCMs, we will see that the dominant processes for the eddy-driven shift are in need of some modification.

4.1 EP-flux adjustment

The climatology of the EP-flux simulations (Fig. 5a) shows typical upward propagation in the lower to middle troposphere, indicating poleward eddy heat fluxes. There is also baroclinic generation, and meridional propagation in the upper troposphere, indicating wave-like propagation and refraction due to sphericity and the curvature in the background zonal wind. The eddy momentum flux convergence associated with the meridional propagation of the wave activity (i.e. horizontal EP-fluxes) maintains the eddy driven jet. During the first 25 days in experiment CAM4_4K, the trend of EP-flux is characterized by (1) a reduction of the vertical flux across the midlatitudes, indicating a reduction in wave production; (2) an enhanced upward and equatorward flux in the upper troposphere that transports more momentum poleward (Fig. 5c), giving rise to a poleward shift of the eddy-driven jet. Both these features bear considerable resemblance to the transient response to a 4°K uniform warming in an aquaplanet AM2.1 used in Chen et al. (2013). There is considerable resemblance in the upper level EP-flux and divergence trends (Fig. 6b) and the associated trend in the convergence of eddy momentum flux (Fig. 6c) in AM21_A1B; whereas the feature of anomalous downward EP-flux trend is absent, likely due to the spatial gradient component in the A1B-type SST perturbation. Examining the first 5 days of the adjustment in CAM4_4K, we instead only see an overall enhancement of the climatological pattern of the EP-flux, together with an enhancement of the mean eddy momentum convergence at the equatorward flank of the mean jet, but no sign of shift at all (not shown). This seems to be a reaction to the initial shock of the planetary boundary layer instability due to the abrupt introduction of SST warming. As the atmospheric column has adjusted to the surface warming and becomes more stratified, the increased static stability acts to suppress the baroclinic growth (Fig. 5b, see also Lu et al. 2008, 2010). It is important to stress that the trend in the vertical EP-flux during the first 25 days does not exhibit any significant meridional structure. Thus, the dipole of the eddy momentum flux convergence/divergence can not be attributed directly to the change of the baroclinicity as the non-acceleration theorem would suggest (Lu et al. 2010; Tandon et al. 2013). A question arises as to what then has happened during the period between day 5 and 25 that leads to the poleward shift of the eddy momentum forcing of the jet.

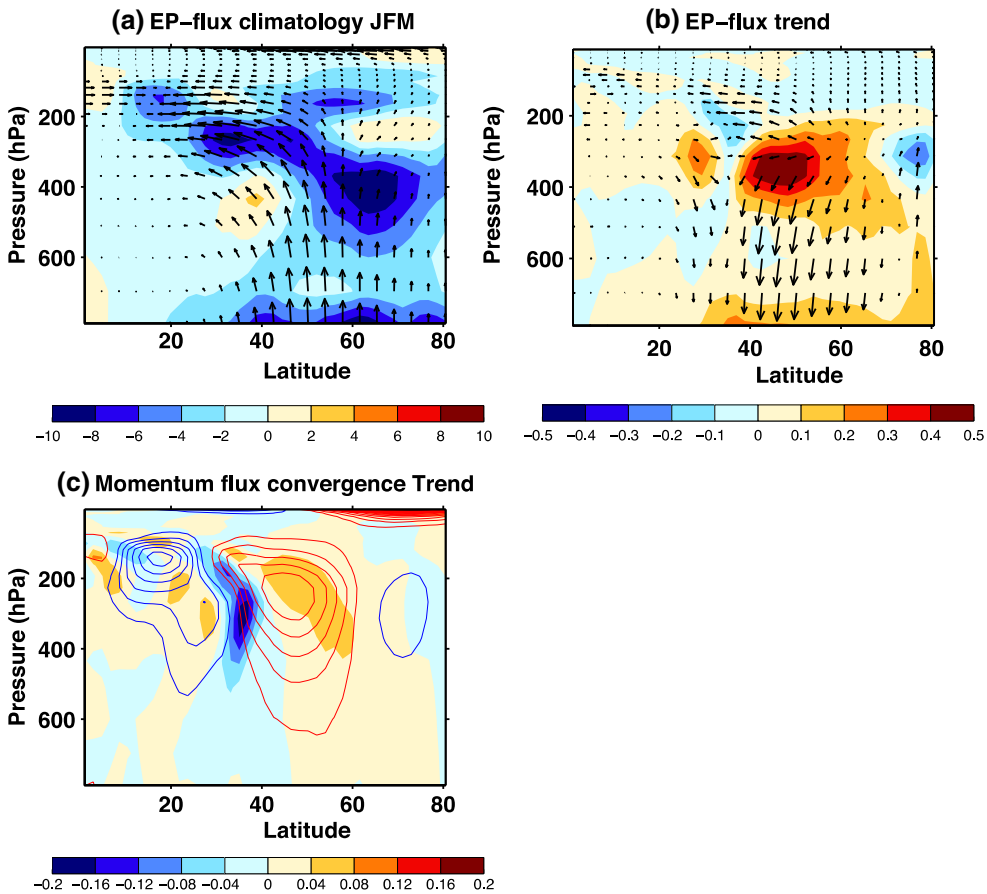

Fig. 5 **a** Climatological EP-flux during JFM; **b** the trend of EP-flux during day 1 to day 25. The *shadings* in **a** and **b** denote the EP-flux divergence (unit is ms^{-1} day^{-1}), the *arrows* the EP flux vectors; **c** eddy momentum flux convergence with the *shading* denoting the 1–25 day trend (unit is ms^{-1} day^{-1}) and the contours the JFM climatology with *red* for convergence and *blue* for divergence, and contour intervals of 1 ms^{-1} day^{-1}. This is the NH result from the CAM4_4K ensemble runs and the corresponding control ensemble CAM4_CNTRL

4.2 Finite amplitude wave activity budget

To address the question regarding the origin of the eddy momentum forcing of the jet shift, following Chen et al (2013) and Lu et al (2014) we perform a QGPV-based FAWA budget onto the transient large ensemble simulations CAM4_4K and AM21_A1B. Despite the different SST perturbations used and differing low-level EP-fluxes in these two experiments, the budget reveals many robust features in common between the two in the upper-level wave activity dissipation.

Figure 7 shows the budget terms for trends during the adjustment from day 1 to 25 over the NH in CAM4_4K ensemble. Following Eq. (4), the trend of $\overline{v'q'}$ (Fig. 7a) is balanced by (1) negative wave activity tendency ($-\partial A/\partial t$, Fig. 7b); (2) the change of diabatic wave activity source ($\Delta\Sigma$, Fig. 7f); and (3) change in the dissipation (Fig. 7c), which can be further broken into the components due to a change in PV gradient (Fig. 7d) and a change in effective diffusivity (Fig. 7e).

The wave activity adjustment towards a lesser total wave activity with SST warming in Fig. 7b is the result of the sum of (1) conversion to zonal momentum by $\overline{v'q'}$ (Fig. 7a), (2) the reduced baroclinic wave production from the lower troposphere (Fig. 5b), and (3) the enhanced dissipation of the wave activity near the subtropics and high latitudes (Fig. 7c) despite the increased diabatic production just below the mean tropopause level (Fig. 7f). The objective of the budget is to explain the dipole in the PV flux that anchors the upward and equatorward wave activity flux from the positive center (near 40°–60°N, 200–500 hPa) to the negative center (near 30 N, 200 hPa). The associated poleward transport of eddy momentum directly drives the poleward shift of the jet. The smooth wave activity tendency term shows no strong sign of a dipole needed to anchor an anomalous equatorward eddy momentum flux (Fig. 5b), so should not be the cause for the shift of the eddy-driven jet. The diabatic wave source term (Fig. 7f) is mainly characterized by a poleward and upward shift with a relatively weak, though non-negligible, contribution to

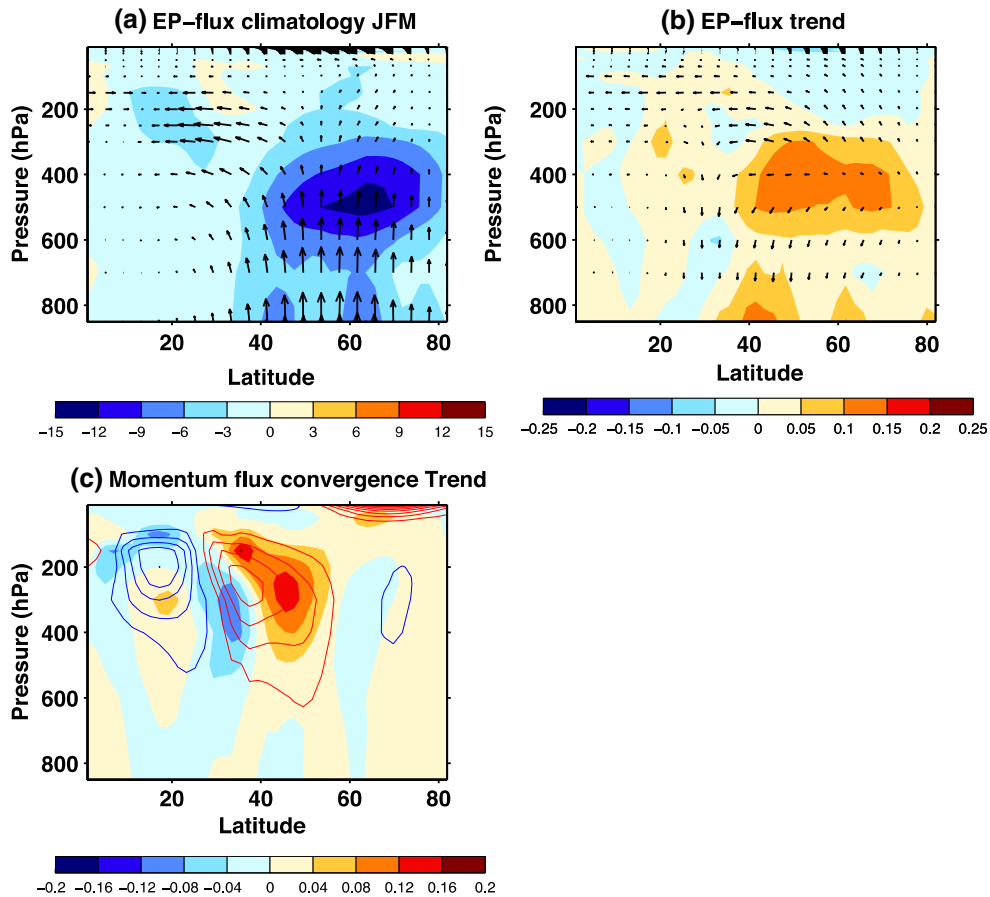

Fig. 6 Same as Fig. 5 except for AM21_A1B ensemble runs and the corresponding control ensemble AM21_CNTRL

the total change of wave activity or $\overline{v'q'}$. The main source of the dipole in $\overline{v'q'}$ arises from the dissipation term, which features a reduced dissipation centered at (45°N, 350 hPa) and an enhanced dissipation at (30°N, 250 hPa), which bear a considerable resemblance to the dissipation dipole found in the response to a tropical upper tropospheric heating simulated by a dry AGCM (Figure 7e in Lu et al. 2014). The large increase of wave activity dissipation at higher latitudes (poleward of the mean jet) seems to be in keeping with the more frequent cyclonic wave breaking at higher latitudes found in both idealized dry model response to a tropical heating (Lu et al. 2014) and the full CM2.1 coupled model response to GHG forcing (Kidston and Vallis 2010). In the latter, the enhanced high-latitude wave breaking was linked to the reduced wave phase speed relative to the background zonal flow and the associated wave breaking at higher critical latitudes at the poleward side of the jet. However, at least in our model experiments, this enhanced wave dissipation seems to be decoupled from the eddy momentum flux (Figs. 5b, 7c) and hence unlikely responsible for the eddy momentum forcing for the jet shift.

Further decomposition of this dipole indicates that it mainly results from the change of effective diffusivity instead of the change in $\frac{\partial Q}{\partial y}$ (compare Fig. 7d and e). This result, despite the added complexity due to the full physics and realistic configuration, corroborates the notion from more idealized model studies (Chen et al. 2013; Sun et al. 2013; Lu et al. 2014) that irreversible mixing of PV and dissipation of wave activity are essential for the origin of the eddy momentum forcing for the jet shift. A subtle difference here from the idealized model studies is that the reduced dissipation (Fig. 7d) due to the weakening of $\frac{\partial Q}{\partial y}$ (Fig. 7g) seems to play a comparable role to the change of K_{eff} in accounting for the total increase of $\overline{v'q'}$. Thus, both the changes in effective diffusivity and in PV gradient are responsible for more wave activity surviving to get to the propagation level in midlatitudes.

The main features in the change of $\overline{v'q'}$ in the upper troposphere are characterized by a tripole with an increase near 40°–45°N flanked by two decreases on both sides. The budget analysis above suggests these features can be traced back to the change of the dissipation of wave activity, which itself is shaped by the structure of the change of effective diffusivity (Fig. 7h). During NH winter, the distribution of the mean effective diffusivity (contours in Fig. 7h) is characterized by a minimum at the subtropics near the winter subtropical jet and large values at its both equatorward and

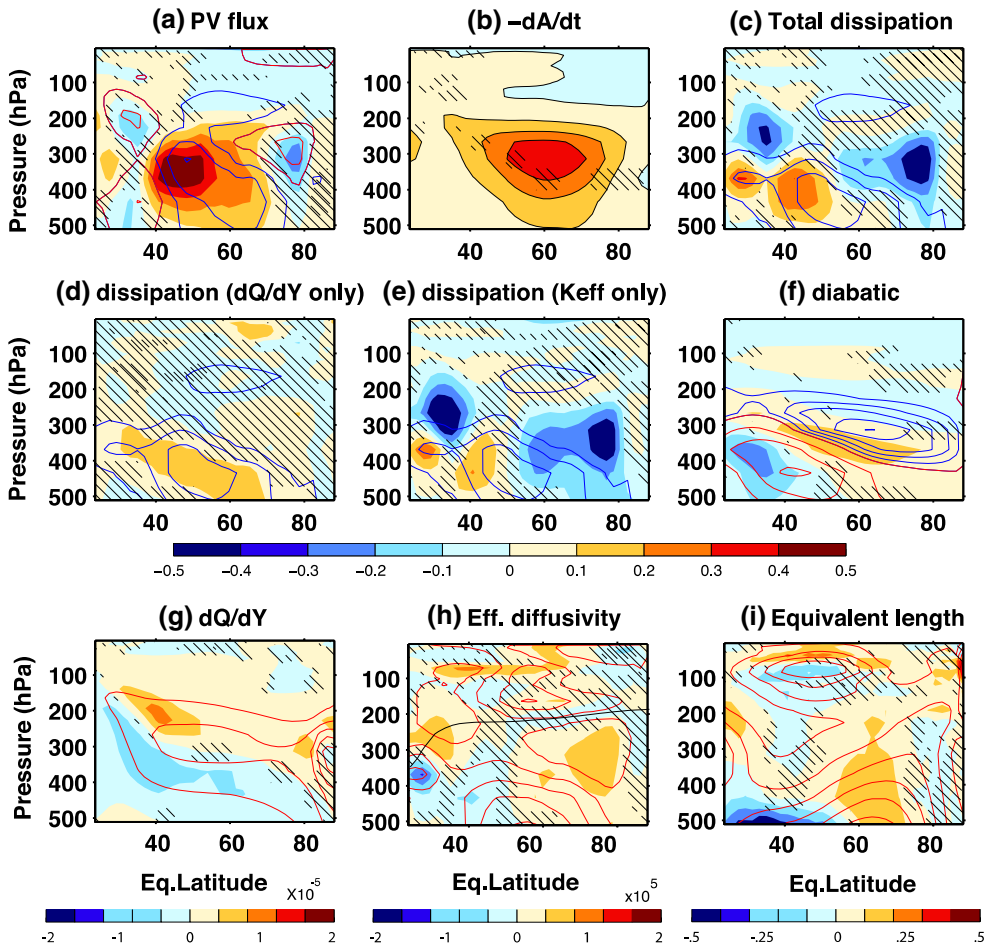

Fig. 7 The linear trend of the wave activity budget for the NH during days 1–25 in CAM4_4K transient ensemble runs. **a** $\overline{v'q'}$; **b** $-dA/dt$; **c** $-\frac{K_{eff}}{a}\frac{\partial Q}{\partial \phi_e}$; **d** the trend in $-\frac{K_{eff}}{a}\frac{\partial Q}{\partial \phi_e}$ due to the change of $\frac{\partial Q}{\partial \phi_e}$; **e** the trend in $-\frac{K_{eff}}{a}\frac{\partial Q}{\partial \phi_e}$ due to the change of K_{eff}; **f** $\Delta\Sigma$; **g** $\frac{\partial Q}{\partial \phi_e}$; **h** K_{eff} derived from residual approach; and **i** equivalent length from direct calculation. The contours in **a**, **f**, **g**, **h** and **i** are the background JFM mean $\overline{v'q'}$ (CI = 3 ms^{-1} day^{-1}), $\Delta\Sigma$ (CI = 3 ms^{-1} day^{-1}), $\frac{\partial Q}{\partial \phi_e}$ (CI = 2.5 × 10^{-4} s^{-1}), K_{eff} (CI = 1 × 10^6 m^2 s^{-1}), and normalized equivalent length (unitless, CI = 5), respectively; and contours in **c**, **d**, **e** are the JFM climatological mean $-\frac{K_{eff}}{a}\frac{\partial Q}{\partial \phi_e}$ (CI = 4 ms^{-1} day^{-1}). The *hatched areas* are where the trend is insignificant at the 5 % confidence level using Student's *t* test. The *black line* in **h** marks the pressure level of 330 K isentrope

poleward flanks. The trend in the upper troposphere acts to enhance (reduce) the effective diffusivity near upper (lower) portion of the mean subtropical maximum, thus shift the total maximum upward. In the meantime, the minimum of K_{eff} tends to shift poleward during days 1–25. In the mid- and high-latitudes, K_{eff} increases over a broad region.

A cautionary note should be added that this residual estimate of K_{eff} is not expected to be accurate given the many assumptions made in the budget Eq. (4), especially the QG approximations. As an *a posteriori* validation, we compute the normalized equivalent length, which represents K_{eff} were k constant, using formula $\frac{|\nabla q|^2}{\frac{\partial Q}{\partial y}^2}$. The result (Fig. 7i), in a sense of relative change with respect to its own mean state, appears to confirm the main characters of the trend of K_{eff}. It follows that the insights gained from the wave activity budget above is likely to hold for the response in question, not susceptible to errors due to the QG approximations.

Almost all the features described above (except those about the diabatic wave activity source, which is not available for AM2.1 model), are also found in the transient evolution in AM21_A1B (see Fig. 8). In particular, the tripolar structure of the K_{eff} change is also present, suggesting the robustness of this feature. The major difference from its CAM4 counterparts is the stronger cancellation in the total dissipation between the part due to the change of K_{eff} and that due to the change of $\frac{\partial Q}{\partial y}$.

4.3 PV gradient change

An alternative approach for understanding the jet shift is through the evolution of the upper level PV gradient, considering the constant relationship between its maximum and the jet. Butler et al. (2011) painted a simple and compelling picture about the eddy adjustment in response to an imposed tropical heating: the upper tropospheric warming drives a dipolar structure in the Ertel PV gradient near the tropopause, and eddies respond with a straightforward down-gradient PV flux, which in turn drives the shift of the eddy-driven jet. Here, we examine the eddy PV flux, PV gradient and its diabatic source in our transient ensemble experiments with two full AGCMs and compare with the depiction of Butler et al. (2011). Note that in their case, the anomalous PV gradient dipole is broad, and straddles the mean jet, with an increase at the poleward flank and a decrease at the equatorward flank.

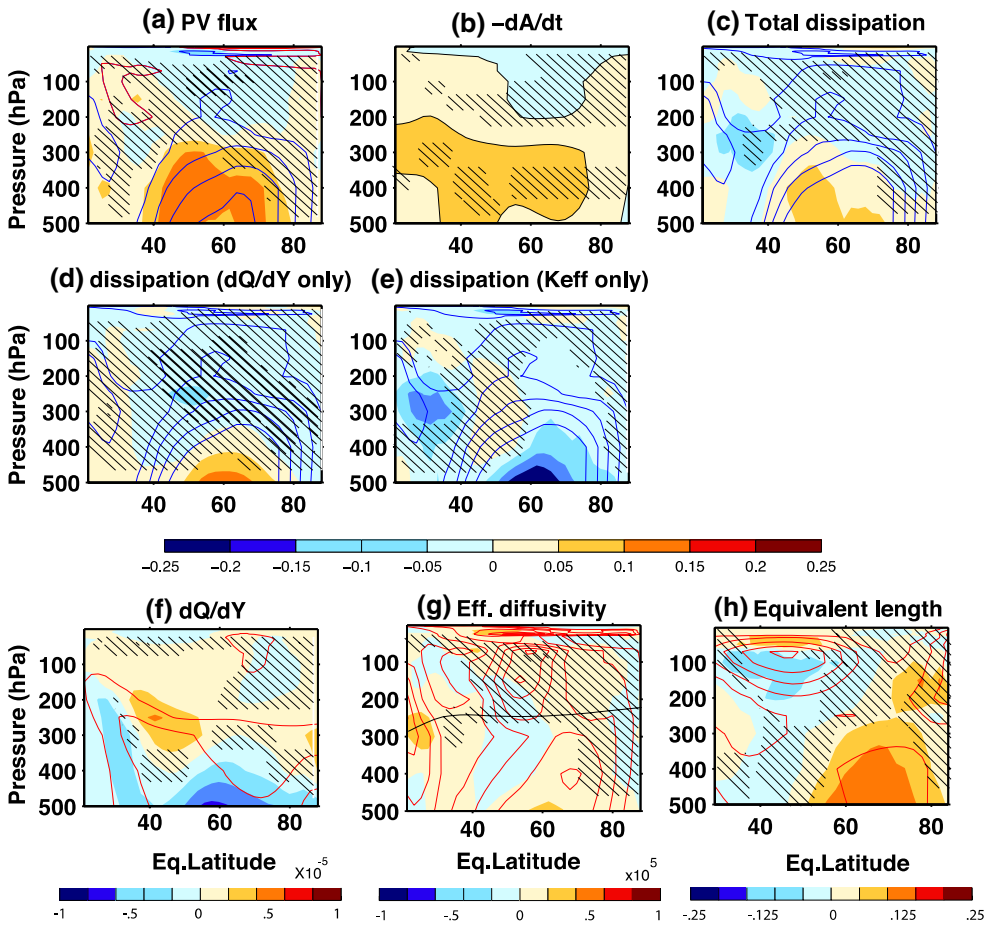

Fig. 8 Same as Fig. 7 but for the AM2.1 simulations. Note that since the diabatic heating is not available for the AM2.1 simulations, the panel for the diabatic wave source term has to be omitted. The *black line* in **g** marks the pressure level of 330 K isentrope

To facilitate the comparison with Butler et al. (2011) as well as Lu et al (2014), we have interpolated the QG PV from isobaric coordinate onto isentropic surfaces. This approach can be justified by the fact that the meridional eddy QG PV flux is a good approximation for the residual eddy Ertel PV flux in the QG limit on isentropic surfaces (Plumb and Ferrari 2005). In both CAM4_4K and AM21_A1B ensembles, the anomalous PV gradient (contours in Figs. 9a, 10a) exhibits a dipolar structure from day 5 (remember that jet has not yet shifted at day 5), similar to Figure 7a in Butler et al. (2011) and Figure 8a in Lu et al. (2014). However, the extrema of the PV gradient here are fixed in space in our moist cases, with little poleward migration with time as seen in those dry simulations. Moreover, the negative pole of the dipole here is much more confined meridionally, compared to that in Lu et al. (2014) and Butler et al. (2011), likely a result of different mean states. In terms of the relative phase relationship between the eddy PV flux and the PV gradient, our full model simulations agree much more closely with Lu et al. (2014) than with Butler et al. (2011), that is, the meridional phase of the anomalous eddy PV flux is roughly in quadrature with that of the PV gradient while they are out of phase with each other in Butler et al. (2011). Admittedly, this might not be a fair comparison as the PV gradient in Butler et al. (2011) is based on Eulerian mean, while both Lu et al. (2014) and this study use contour-following, Lagrangian mean PV gradient. In addition, as argued in Lu et al. (2014), the eddy PV flux is mainly maintained by the change of the PV diffusion due to the change of effective diffusivity (rather than the PV gradient per se; see contours in Fig. 9b), as can be inferred from the their spatial and temporal similarity (comparing the shadings in Fig. 9a, b). The same can be concluded about the relationships among eddy PV flux, PV gradient, and diffusion of PV by change of effective diffusivity revealed by the AM21_A1B ensemble (see Fig. 10), despite the different SST forcing therein. Taken together, the eddy adjustment processes depicted in Lu et al (2014), which underscores the importance of the change in effective diffusivity, may be more representative of what might be happening in the real atmosphere in response to the GHG-induced tropical warming.

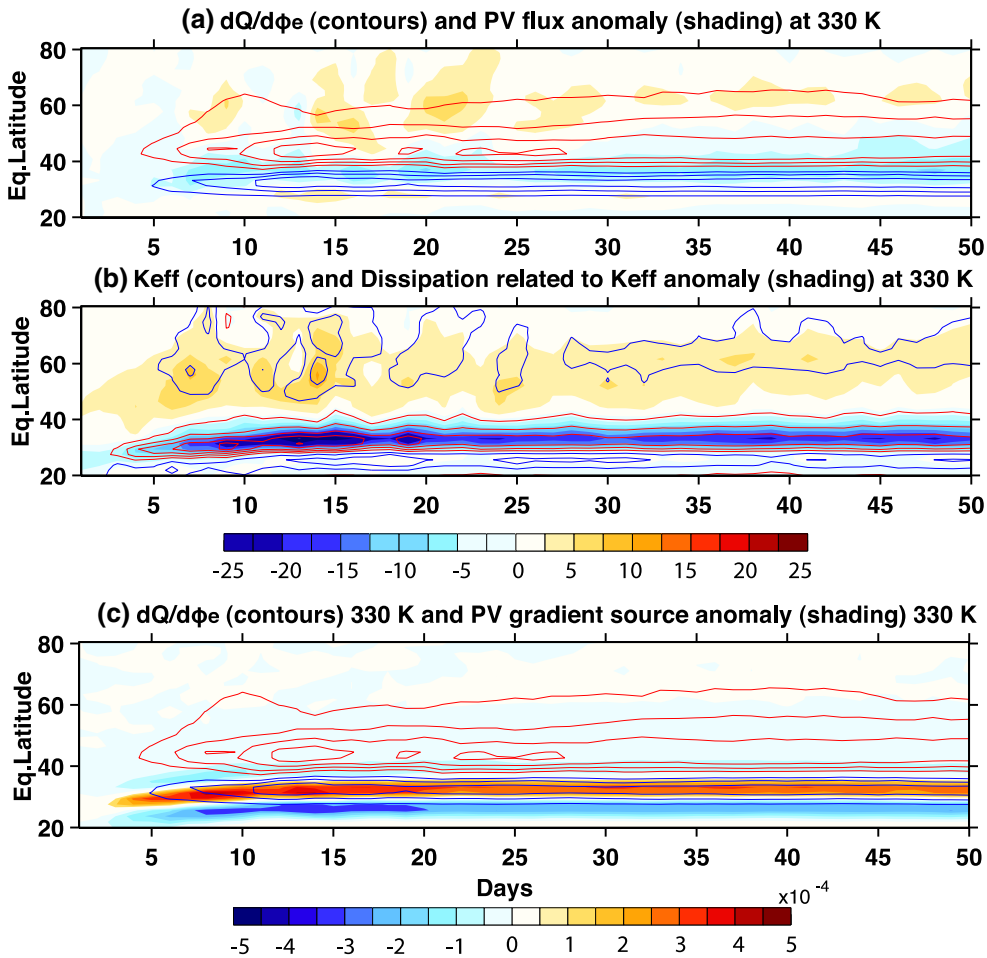

Fig. 9 Temporal evolution of the ensemble mean response to 4 K uniform warming in CAM4. **a** *Shading* indicates the response of the PV flux at 330 K isentrope (CI = ms^{-1} day^{-1}) and contours the corresponding PV gradient (CI = 6×10^{-5} s^{-1}); **b** *shading* indicates the response of the component of wave activity dissipation due to the change of K_{eff} at 330 K (CI = ms^{-1} day^{-1}) and contours K_{eff} at 330 K (CI = 4×10^5 m^2 s^{-1}) from the residual approach. **c** *Shading* represents the response of the PV gradient source at 330 K (CI = 6×10^{-5} s^{-1}) and contours the same PV gradient as in **a**

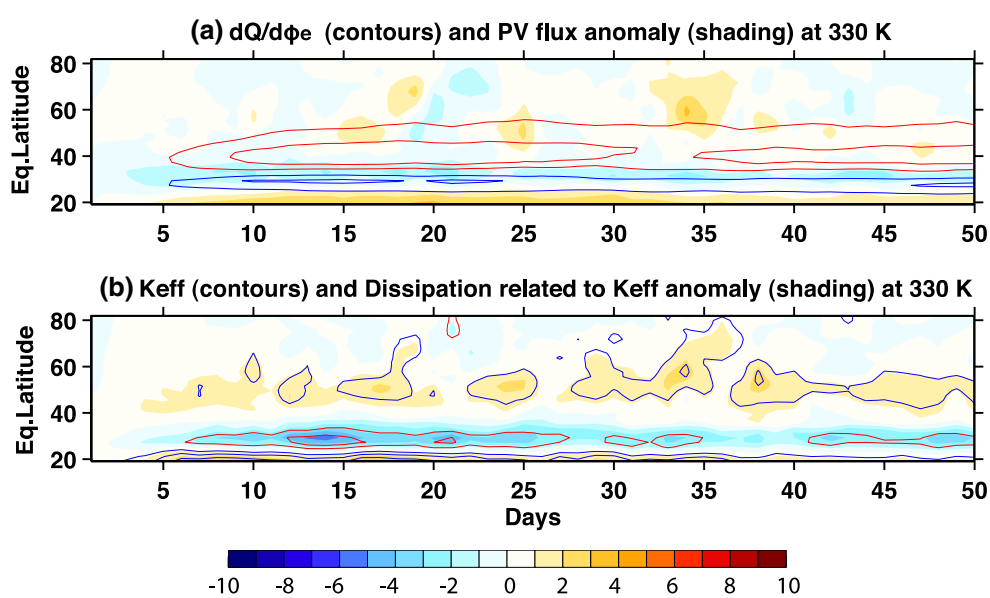

Fig. 10 Same as Fig. 9a, b, but for the response to A1B SST warming simulated in AM2.1

We further diagnose the diabatic source for the instigation and evolution of the Lagrangian PV gradient, as it can only be changed by diabatic and other non-conservative processes. Following Nakamura and Zhu (2010, their Eq. 48), we can express the budget for $\partial Q/\partial y$ as

$$\frac{\partial}{\partial t}\frac{\partial Q}{\partial y} = \frac{\partial^2}{\partial y^2}\left(K_{eff}\frac{\partial Q}{\partial y}\right) - \frac{\partial^2}{\partial y^2}\Delta\Sigma - \frac{\partial^2}{\partial y^2}\bar{X}, \tag{6}$$

where the first term on the right hand side represents the irreversible PV diffusion, the second term the diabatic source, and the third term friction, which is usually small compared to the other terms in the free atmosphere. Thus, the balance of Eq. (6) is largely maintained among the tendency, diffusion, and diabatic source terms. In Fig. 9c we present the time evolution of $\frac{\partial Q}{\partial y}$ and its diabatic source term interpolated onto the 330 K isentrope. The upshot of this analysis is that diabatic heating does not drive the PV gradient change directly, rather mostly opposes the tendency of the PV gradient during its emerging phase. By corollary, it has to be the irreversible mixing of PV through wave breaking that is directly responsible for the change of PV as evidenced in Fig. 9b.

4.4 Mechanism for the cross-jet wave propagation

The wave activity budgets of the wave response in models across a spectrum of complexity highlights the importance of the reduction of wave activity dissipation under SST or diabatic warming associated with GHG warming. Further questions naturally arise as to why the surviving wave activities propagate equatorward across the mean jet in the upper troposphere/lower stratosphere and when this first takes place in the transient adjustment.

A trivial answer to the first question would be that this anomalous wave activity just follows the prevalent pathway of the mean wave activity propagation. But closer inspection of the eddy momentum flux in our spinup experiments (the meridional components in Figs. 5a, b, also SH) indicates that the anomalous flux appears to emanate from relatively higher latitude than the climatological one. Recently, through the development of Rossby wave chromatography, Lorenz (2014a, b) advocated the importance of "reflecting level" dynamics for the positive feedback involved in the poleward shift of the jet under several thermal and mechanical perturbations to the Held and Suarez (1994) dry dynamical core. More specifically, "when the zonal wind increases on the poleward flank of the jet, a wider range of poleward propagating waves encounter a reflecting level instead of a critical level on the poleward flank." The increased wave reflection leads to increased equatorward propagation of waves across the jet, a ubiquitous signature found in many simulations for the jet shift (Figures 3–4 in Lorenz 2014a; Figure 6 in Sun et al. 2013; Figure 9 in Chen et al. 2013). To test whether the signatures of Lorenz (2014b) are also present in the eddy momentum flux anomalies from our spinup simulations, we follow Lorenz (2014b) and adopt the dispersion relation for a linear, unforced, barotropic plane wave perturbation on background zonal mean wind $\bar{u}$ in Mercator coordinates for describing the angular phase speed:

$$c_\omega = \frac{\bar{u}}{acos\phi} - \frac{\beta^* acos\phi}{m^2}, \tag{7}$$

where β^* is the zonal mean absolute vorticity gradient, m is the integer zonal wavenumber and c_ω is the angular phase speed. In deriving (7), the meridional wave number l is set to be 0 in the dispersion relation. So c_ω represents the phase speed level at which $l = 0$, or reflection occurs. At a given latitude, waves propagating poleward from their mid-latitude source with phase speed less than c_ω are reflected toward the equator.

The c_ω profiles at 267 hPa level for the CAM4_CNTRL (averaged from day 1 to day 25) and control plus the linear trend during that period in CAM4_4K ensemble are presented as the black and red dash lines in Fig. 11. Also plotted are the zonal mean zonal wind and the trend of meridional EP-flux. The 267 hPa level is chosen because it represents the center of maximum of the meridional EP-flux as well as the pressure range where the change of the dissipation of wave activity is relatively small and wave-like behavior may be more potent there. As suggested by Lorenz (2014b), the reflection phase speed indeed increases at the poleward side of the mean jet, and accordingly more waves are reflected from there across the jet to be finally dissipated near the subtropical critical latitudes, giving rise to an anomalous poleward transport of momentum from the subtropics to the poleward side of the jet. The main characteristics of Fig. 11 agree with the schematic of Lorenz (2014b, his Figure 4c), although the profiles of the reflecting phase speed in our case are somewhat different from those in Lorenz (2014b). In our case, the profiles of reflecting level peaks at ~40°N as it is dominated by the curvature in the strong winter subtropical jet, while in Lorenz (2014b) the reflecting level peaks at higher latitude as it is dominated by the curvature in the eddy-driven jet (see Figure 3 in Lorenz 2014b). Note that Fig. 11 shows results for fixed zonal wave number (at 6), hence it does not take into account the increase of the wave length, which is a robust change in the mid-latitude waves in response to global warming found in CMIP3 and CMIP5 (Kidston et al. 2010; Barnes and Polvani 2013). Following Eq. (7), increasing wavelength alone would decrease the reflecting phase speed. Thus, the increase here can only be attributed to the change of the profile of the background zonal wind. Staten et al. (2014) performed a similar analysis on the

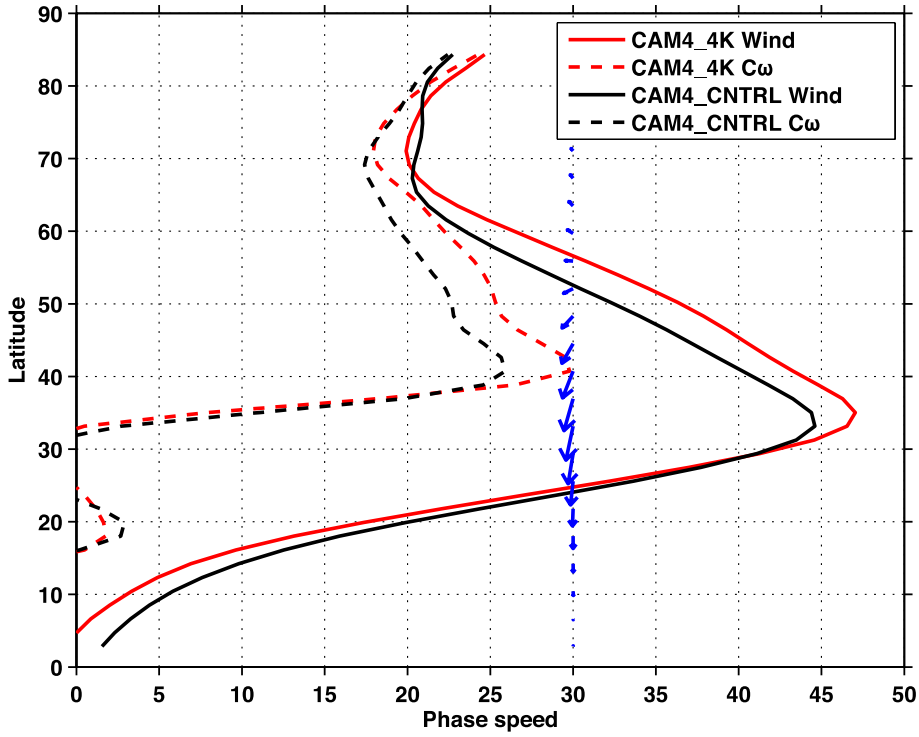

Fig. 11 Reflecting angular phase speed (C_ω, *dashed*, unit is m/s) plotted together with the zonal mean zonal wind divided by *cosϕ* (*solid lines*, unit is m/s), and the trend in the EP flux during days 1–25 (*arrows*) at 250 hPa level. The *black lines* are for the background mean profiles from CAM4_CNTRL and the *red lines* for the total profiles at day 25 simulated in CAM4_SST. The *leftward pointing arrows* represent upward EP-flux

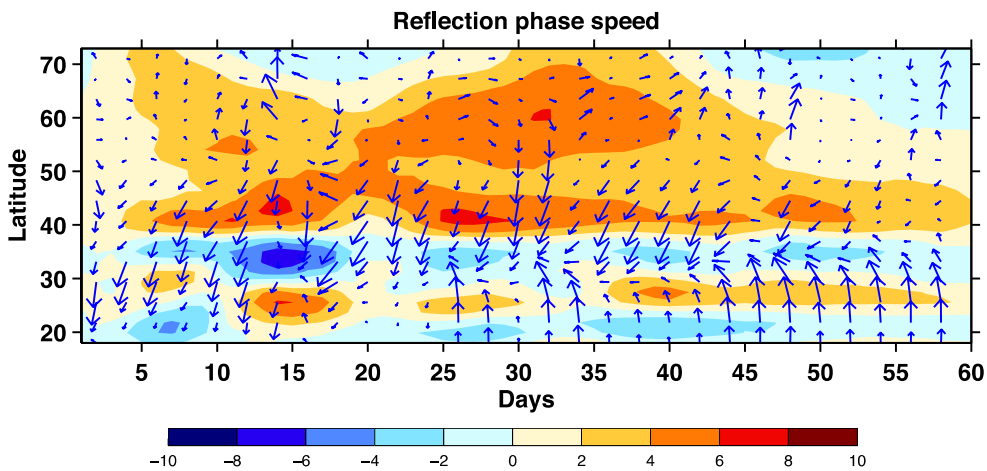

Fig. 12 Temporal evolution of the anomalous reflection phase speed C_ω (*shading*) and EP-flux (*arrows*) at 267 hPa in response to 4 K uniform warming in CAM4_SST-CAM4_CNTRL. The *leftward pointing arrows* represent upward EP-flux. Note that a 5-day running mean has been applied to highlight the smooth features of the adjustment

spinup experiment of AM21_A1B and found a strong relationship between the increase of the reflection level phase speed and the shift of the jet, supporting the proposition of Lorenz (2014b) on the positive feedback through enhanced reflection in the jet shift. The same feedback mechanism appears to operate in the CAM4_4K experiment as well.

It remains to confirm whether the increase of reflection phase speed initiates the chain of events that lead up to the poleward shift of the jet, as advocated by Lorenz (2014a, b). To this end, we examine the full evolution of the reflection phase speed anomalies together with the EP-flux anomalies at 267 hPa level (Fig. 12). Throughout the temporal adjustment, the reflection phase speed increases poleward of 40°N and decreases between 30°N and 40°N, directing wave activity equatorward across the mean jet and producing momentum convergence at the poleward side of the mean jet. This process happens at the very early stage of the adjustment and reaches its first peak at approximately day 12, producing the first momentum burst (cf. Fig. 4a). However, this process takes place simultaneously with the

occurrence and poleward advance of the K_{eff} anomalies and one cannot discern which one occurs earlier than the other (cf. Fig. 9b) so as to serve as an candidate for the initiator for the shift of the eddy-driven jet. On the other hand, a greater equatorward propagation of wave activity should lead to enhanced wave breaking at the equatorward flank of the jet and hence a greater effective diffusivity there. This seems to be exactly what is happening, explaining partly the initial increase of K_{eff} at the subtropical latitudes. At the later stage (after day 25), a second positive anomaly of reflection phase speed appears at the equatorward side of the subtropical jet around 25°N (resulting from the change of the curvature of the subtropical jet), reflecting waves poleward. These waves collide with those reflected from higher latitudes at around 35°N, maintaining the band of K_{eff} increase nearby.

The increase of the reflection phase speed in the mid-to-high latitude band is reconcilable with the reduction of the effective diffusivity (Fig. 9b, contours). An increase in the equatorward reflection from the turning latitude would imply less wave breaking and less effective diffusivity for the dissipation of wave activity at the poleward flank of the mean jet. Nevertheless, the decrease of effective diffusivity in the midlatitudes (40°–60°N in Fig. 7h and Fig. 8g) is more readily explained by the increase of the zonal wind as the zonal wind acting as a mixing barrier is well established in the studies of chaotic mixing and transport by quasi-geostrophic flows (e.g., Pierrehumbert and Yang 1993; Nakamura 1996; Haynes and Shuckburgh 2000; Marshall et al. 2006).

4.5 Southern hemisphere response

Turning to the Southern Hemisphere, since the equilibrating time of the SH wind is much longer than the NH, we will be focusing on the trend during the first 50 days in the switch-on ensembles. In view of the notable difference even in the climatological mean zonal wind between the two models (compare the contours in Figs. 2a and 3a) and the fact that the trend part also exhibits some qualitative differences between AM21_SST and CAM4_SST in the SH, the representation of the eddy adjustment process in the SH is less certain. Thus, we only focus on features shared in common between the trends in AM2.1 and CAM4.

Figures 13 and 14 show the JFM mean EP-flux and eddy momentum fluxes and the corresponding trends during the first 50 days of the CAM4 and AM2.1 experiments. In keeping with the different mean winds, substantial

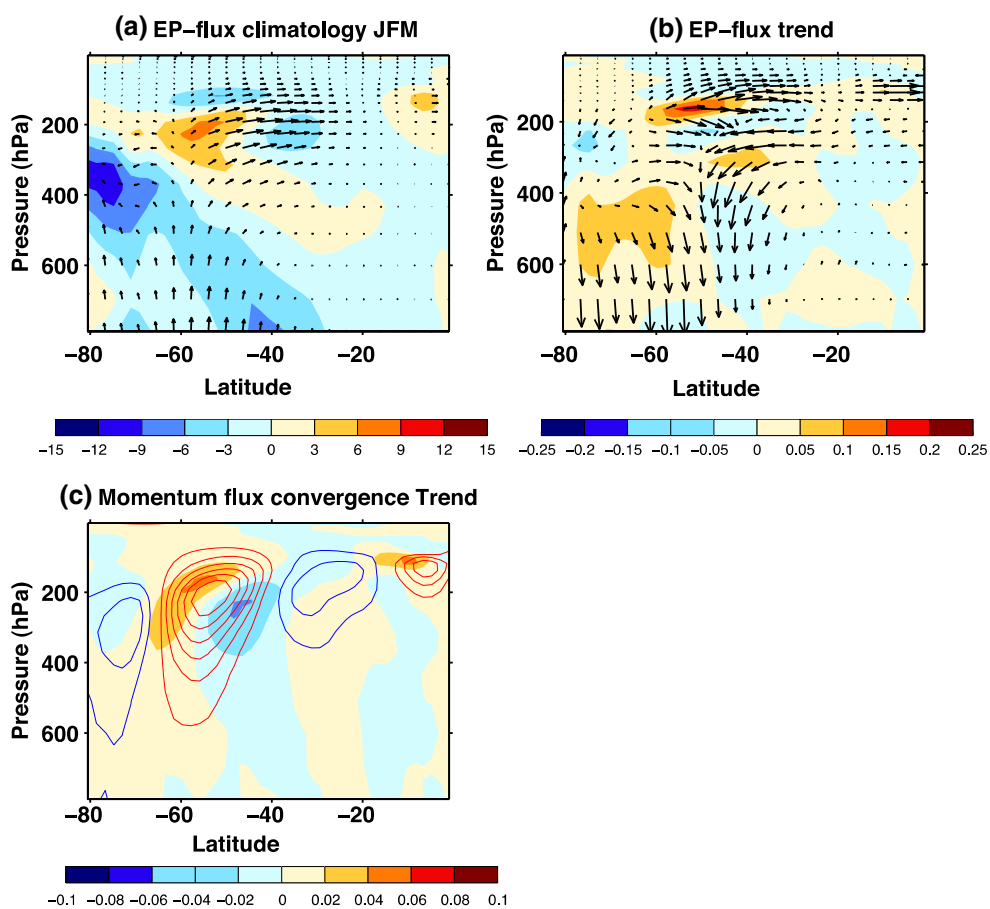

Fig. 13 Same as Fig. 5 except for the SH and for the trends during the period day 1-day 50 in **b** and **c**

differences can also be found in the climatological EP-flux between the two models, with the EP-flux divergence/convergence being more concentrated at upper levels around the tropopause in CAM4, a feature not present in the ERA40 reanalysis (not shown). Correspondingly, the associated eddy momentum forcing for the jet shift is also accentuated higher in altitude in CAM4. In view of the peculiar features of the SH EP-flux in CAM4, we should only focus the SH EP-flux response to SST warming in the AM21 experiments. The results from CAM4 are displayed (Fig. 13) only for the purpose of reference.

By a cursory inspection on Fig. 14b, one notices considerable similarity between the trends of the EP-flux response between the SH and NH (compare Fig. 14b against Fig. 6b). Indeed, most of the descriptions about the wave activity adjustment in the NH can be applied to the SH adjustment as well. For the purpose of brevity, we here do not show the details of the wave activity budget analysis for the SH, but only reiterate the main features consistent with its NH counterpart and point out a few apparent differences.

As for the features that are shared in common with the NH adjustment, (1) the vertical EP-flux is reduced in the lower troposphere, likely the consequence of the increased static stability; (2) the midlatitude wave activity adjusts towards a new equilibrium state with a weaker magnitude, during which time wave activity is being converted to zonal wind at higher latitudes (the corresponding $-\frac{dA}{dt}$ mirrors that in Fig. 8b); (3) the structure in the $\overline{v'q'}$ trend comes mostly from the change of the dissipation of the wave activity, the spatial pattern of which is, in turn, dominated by the contribution from the change of the effective diffusivity; and (4) the change of PV gradient also makes a notable contribution to the reduced dissipation, but does not play as an important role in anchoring the equatorward component of the EP-flux (or poleward flux of momentum).

As for the differences, (1) the locations of the centers of these features, which are all poleward displaced in the SH response compared to their NH counterparts, consistent with the fact that the mean jet in the SH is more poleward than NH during JFM months; (2) there is a strong reduction of wave activity dissipation in the midlatitude stratosphere at the lower portion of the "surf zone". This feature is even more pronounced in CAM4 and accounts for most of the strong positive EP-flux trend in Fig. 13b at about 170 hPa

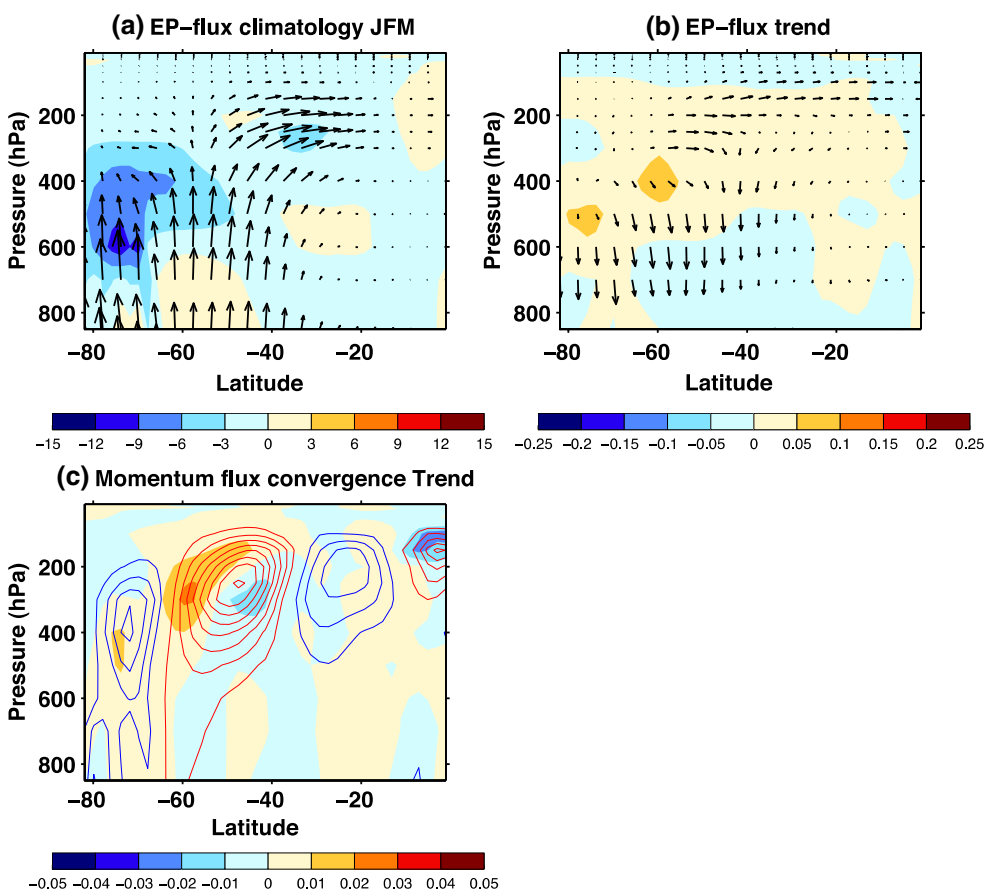

Fig. 14 Same as Fig. 6 except for the SH and for the trends during the period day 1-day 50 in **b** and **c**

between 40° and 60°S, a feature appearing to play more important role in the meridional momentum transport than those in the troposphere. However, due to the coarse vertical resolutions of both AGCMs used here, we can not place much confidence on this feature and defer its investigation until when similar experiments with better resolved stratosphere are available.

5 Summary

To understand the zonal mean atmospheric circulation response to the climate change due to increasing concentration of greenhouse gases, we conducted a series of large-ensemble simulations using two comprehensive AGCMs. Utilizing two different AGCMs helps to identify the robust circulation patterns and the related dynamical processes independent of model configurations. A hybrid Euler-Lagrangian diagnostic framework for the finite-amplitude wave activity is constructed so that the irreversible mixing effect due to wave breaking can be quantified. This tool is applied to the large-ensemble simulations to identify the robust dynamical aspects during the transient adjustment of the mid-latitude westerly jet towards a poleward shifted state.

The result confirms the earlier notion derived from more idealized AGCM experiments regarding the importance of the irreversible PV mixing and dissipation of wave activity in the poleward shift of the eddy-driven jet under global warming (Chen et al. 2013; Lu et al. 2014). This involves a decrease of effective diffusivity in midlatitudes and an associated reduction in the dissipation of wave activity there, which allows more wave activity to survive the mid-latitude dissipation. This also involves an increase of effective diffusivity in the upper-tropospheric subtropics, which acts to homogenize the upper tropospheric PV there, pushing the maximum PV gradient poleward. The structure in the change of effective diffusivity appears to dominate the structure in the change of the eddy QG PV flux, and hence that of the eddy momentum flux convergence. The changing diabatic heating plays a secondary and even opposing role in the formation of the PV gradient in response to a sudden switch-on of SST forcing. The description above seems to be independent of the atmospheric model, model configurations, and to some extent, even the details of the SST forcings. On the other hand, in comparison to more idealized studies, the wave activity budget for the NH adjustment in the full AGCMs presented here suggests a greater importance for the wave activity tendency and the effect of the changing PV gradient on the dissipation of the wave activity.

It is not guaranteed that an enhanced upward wave activity flux (or EP-flux) will inevitably lead to an equatorward cross-jet propagation of wave activity, the direction of which depends on the relative position of the anomalous upward wave activity flux and the change of the upper level refractivity. For the NH wave adjustment during winter months (JFM) in our full AGCM experiments, the enhancement of the upward wave activity flux mostly occurs at the equatorward side of the increase of the reflecting phase speed, and more waves are thus refracted equatorward, resulting in an overall increase in the equatorward cross-jet propagation and more wave breakings near the subtropics, and hence a greater effective diffusivity there. Although the cross-jet propagation is a recurrent feature in the eddy response to a variety of thermal and mechanistic perturbations and appears to be consistent with what the Rossby wave chromatography of Lorenz (2014a, b) depicted, it remains to see whether it originates from linear wave reflection/refraction, or nonlinear wave breaking.

Acknowledgments E.P. and E.K.S. are supported by NSF Grant AGS-1064045. J.L. is also partly supported by the Office of Science of the U.S. Department of Energy as part of the Regional and Global Climate Modeling Program. G.C. is supported by NSF Grant ATM-1064079 and DOE Grant DE-FOA-0001036.

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